Quasars with Flare/Eclipse-like Variability Identified in ZTF

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Paper Link

Abstract

Active galactic nuclei (AGNs) are known to exhibit optical/UV variability and most of them can be well modeled by the damped random walks. Physical processes that are not related to the accretion disk, such as tidal disruption events (TDE) or moving foreground dusty clouds, can cause flare-like and eclipse-like features in the optical light curve. Both long-term and high-cadence monitoring are needed to identify such features. By combining the Sloan Digital Sky Survey (SDSS), Panoramic Survey Telescope, and Rapid Response System (Pan-STARRS) with the Zwicky Transient Facility (ZTF) survey, we are able to identify a rare sample (11) out of the SDSS quasar catalog (~ 83 000). These quasars exhibit more or less constant brightness but show rapid optical variation in the ZTF DR2 epochs. To investigate the possible origins of these flare/eclipse-like variabilities, we propose the second epoch spectroscopic observations with the Gran Telescopio CANARIAS (GTC). We find that the change in accretion rate plays a significant role in these quasar variabilities. Among them, we identify two Changing-Look Active Galactic Nuclei (CL-AGN) candidates: SDSS J1427+2930 and SDSS J1420+3757. The luminosity change of the former may be caused by the enhanced SMBH's accretion or the tidal disruption event, while the latter is more related to the change in the accretion rate.

Sample Selection & Observation

Selection criteria

- luminous quasars: $m_r < 20$ mag
- recent significant changing: $\Delta m_r > 0.35$ mag
- historic constant photometry: $\Delta m_{r_{pre}} < 0.15$ mag
- short duration: $\Delta t < 0.5$ yr

New spectra observation

- GTC/OSIRIS long-slit spectra, 540 s.

Finally, 11 quasars are selected and observed. (PID: GTC40-20A, PI: Shuowen Jin)

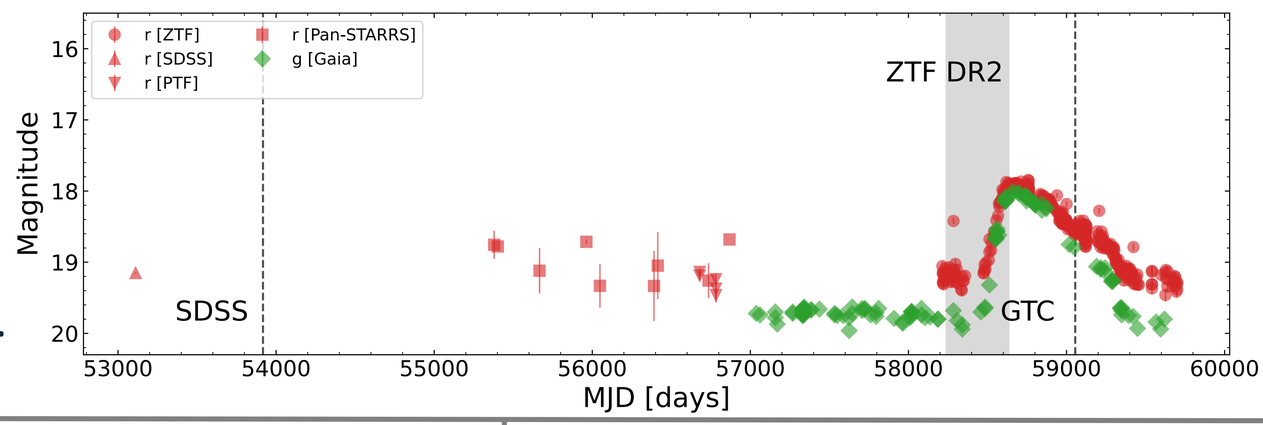


Fig 1: light curve example. The grey shadowed region represents the ZTF DR2 epochs (March 2018 to June 2019), these quasars show rapid luminosity variation in this period. The two dashed vertical lines refer to when SDSS and GTC spectra were observed.

Changing Obscuration ?

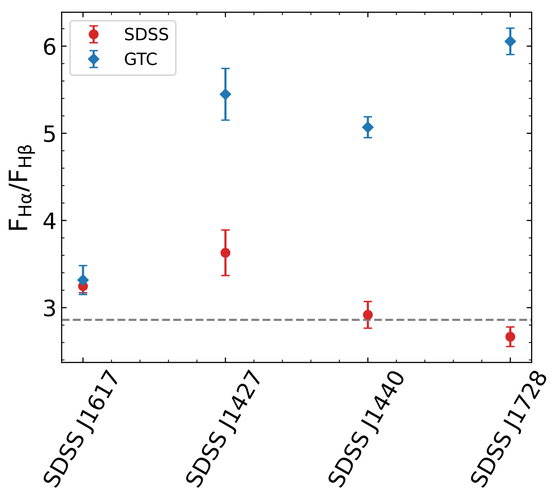


Fig 2: Balmer Decrement. Balmer decrement of 4 quasars with both Ha and Hb detections. the grey dashed line marks the Case B recombination.

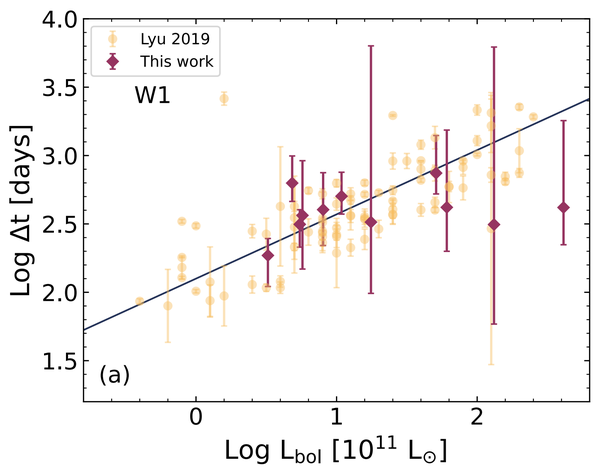


Fig 3: time-lag to bolometric luminosity mid-IR to optical time lag probes the torus size of AGN, the result indicates a normal torus size of our sample compared with PG quasars (Lyu et al. 2019).

Torus size-luminosity relation

- The material from the torus meets the difficult that a normal torus size can not support such a short duration.

Balmer Decrement

- Not only obscuration changes, but also the change in **ionization state.**

The change in obscuration is hard to explain the short duration !

Changing accretion rate ?

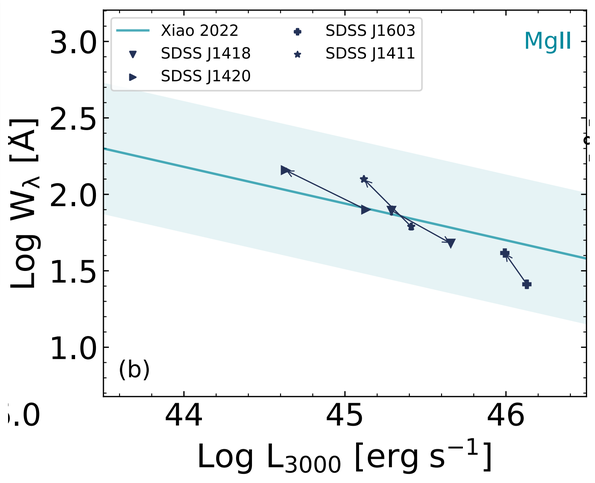


Fig 4: Baldwin Effect. (0.7 < z< 1.9). EW of Mg II broad emission line vs. L_3000 in two epochs, they are all line in the best-fit Baldwin effect within 1 sigma (Xiao et al. 2022).

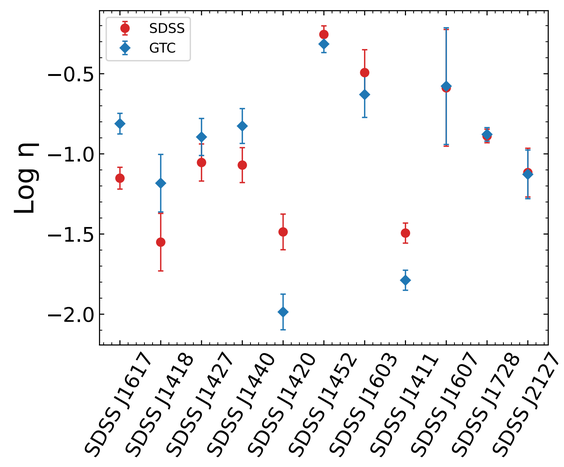


Fig 5: Eddington ratio variation. Eddington ratios comparison between two epochs. Most of them represent significant changes, and consist to the photometry variation.

Baldwin Effect

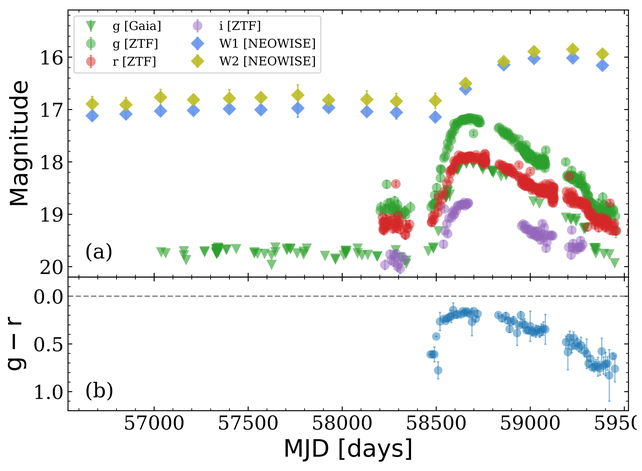
- Inline with Baldwin effect indicating the **ionization state changing.**

Eddington ratio

- The **change in accretion rate** is aligned with the luminosity change in most of our sample. It is also the most possible origin of the ionization state change.

The change in accretion rate plays an important role !

CL-AGN Candidate



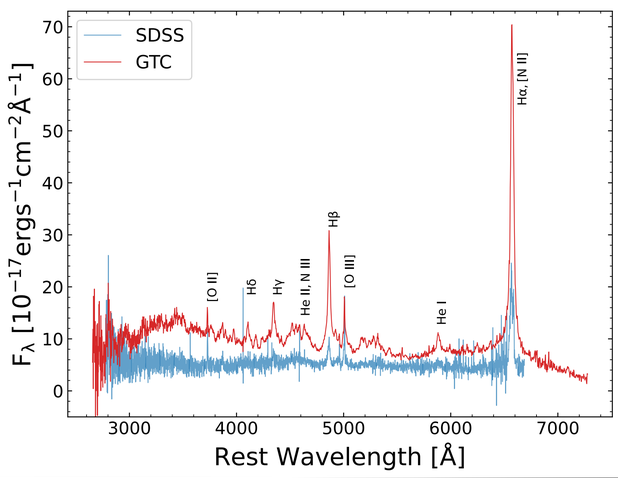


Fig 6: light curve and spectra of SDSS J1427+2930. left: optical and mid-IR light curves, and g-r color evolution. right: SDSS and GTC spectra.

The complex features of light curve and spectra makes it hardly impossible to confirm the origin of this transient.

TDE	AGN Flare
TDE like light curve	Bowen fluorescence
Broad HeII emission	Long decay time-scale
mid-IR echo	Large SMBH mass ($\sim 10^8$ $M_\odot$)

? ? ? ? ?

Take Home Message

- In our sample, four out of the 11 quasars have reliable Balmer decrement measurements, and three quasars show significant changes in the broad-line Balmer decrement. Notably, three out of the 11 quasars show no obvious variation in the Eddington ratio. All 11 quasars in our study exhibit characteristics consistent with a standard torus size and align to the Baldwin effect.
- In line with the Baldwin effect, it is more plausible that the detected variations in the Balmer decrement are attributed to shifts in the ionization state. Additionally, the presence of the Baldwin effect may indicate that **the change in accretion rate plays a substantial role in these flare/eclipse-like AGN variabilities.**
- Following the spectral type diagnosis, we find two CL-AGN candidates, SDSS J1420+3757 and SDSS J1427+2930. The former changes its type from type 1 to type 1.2, while the latter changes from type 1.2 to type 1 between the SDSS and GTC periods.

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